LEXI JOHNSON

Email: alex.johnson@email.com | Phone: (713) 555-0147 | Location: Houston, TX
LinkedIn: linkedin.com/in/alexjohnson-ai | GitHub: github.com/alexjohnson

---

PROFESSIONAL SUMMARY

AI Engineer with 5+ years of expertise in designing and deploying production-grade AI systems, including large language models (LLMs), retrieval-augmented generation (RAG) architectures, and machine learning platforms. Proven track record building scalable AI infrastructure, optimizing neural networks, and leading cross-functional teams to deliver AI solutions that drive measurable business impact. Strong background in full-stack AI development, cloud deployment, and prompt engineering.

CORE COMPETENCIES

- **Languages:** Python, JavaScript, SQL, C++
- **AI/ML Frameworks**: PyTorch, TensorFlow, Keras, Hugging Face Transformers, LangChain, LlamaIndex, Scikit-learn
- **LLM & Generative AI:** OpenAI API, Anthropic API (Claude), Prompt Engineering, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Model Fine-Tuning (LoRA), Prompt Optimization
- **Vector Databases & Storage:** Pinecone, Weaviate, Redis, PostgreSQL with pgvector, Milvus
- **Backend & Deployment:** FastAPI, Flask, Django, Docker, Kubernetes, AWS (Lambda, SageMaker, EC2), Google Cloud Platform (Vertex AI, Cloud Functions)
- **Data & Infrastructure:** ETL pipelines, Apache Spark, BigQuery, Data warehousing, Model monitoring, MLOps
- **Specializations:** Computer Vision, Natural Language Processing, Deep Learning, Transfer Learning, Model Optimization, A/B Testing, Statistical Analysis

---

PROFESSIONAL EXPERIENCE

Senior AI Engineer, TechVision AI Solutions, Houston, TX | *March 2023 – Present*

- Designed and deployed LLM-powered retrieval-augmented generation (RAG) assistants for internal teams, cutting knowledge lookup time by ~35%.
- Built and maintained ML microservices (FastAPI, Docker, Kubernetes) with CI/CD and monitoring, ensuring 99.5%+ uptime.
- Optimized model inference (quantization, batching, caching), reducing latency by up to 40% without degrading quality.

AI/ML Engineer, DataStream Analytics, Austin, TX | *January 2021 – February 2023*

- Built an end-to-end machine learning platform processing 2TB+ of daily data using Apache Spark, reducing data pipeline execution time from 4 hours to 18 minutes through distributed computing optimisation
- Developed and deployed 12+ machine learning models (gradient boosting, neural networks, ensemble methods) for predictive analytics across business domains, achieving an average RMSE improvement of 31% over baseline models
- Designed FastAPI-based REST API for ML model inference serving 100K+ requests daily with 99.5% uptime and <200ms latency, integrated with Docker and Kubernetes for scalable deployment

Machine Learning Engineer, InnovateTech Solutions, Dallas, TX | *July 2019 – December 2020*

- Developed and productionized models for classification, recommendation, and forecasting, improving core metrics by 30%.
- Implemented automated training pipelines for data preprocessing, feature engineering, training, and evaluation on cloud infrastructure.
- Collaborated with product and data teams to translate requirements into measurable ML objectives and KPIs.

Junior AI Engineer, CloudTech Data, Houston, TX | *June 2018 – June 2019*

- Supported development of machine learning models using Python and scikit-learn for customer segmentation and churn prediction, contributing to 3 models deployed to production
- Built data visualisation dashboards using Plotly and Tableau, presenting ML insights to non-technical stakeholders and enabling data-driven business decisions
- Participated in a research project on neural network optimisation techniques, contributing code implementation for 2 novel compression algorithms tested on the CIFAR-10 dataset
- Assisted in cloud infrastructure setup on Google Cloud Platform, implementing cost optimisation strategies that reduced monthly infrastructure spend by 22%

TECHNICAL PROJECTS & ACHIEVEMENTS

LLM-Powered Document Analysis System | 2024

- Engineered RAG system processing 100K+ enterprise documents using semantic chunking and hybrid search (BM25 + vector similarity), reducing document retrieval time from 2 minutes to <500ms
- Implemented prompt optimization pipeline using few-shot learning and chain-of-thought prompting, improving document summarization accuracy by 37%
- Deployed on AWS using Lambda functions and Pinecone vector database, handling 50K+ queries monthly with zero downtime

Predictive Maintenance ML Platform | 2023

- Developed an ensemble machine learning model combining gradient boosting and LSTM networks for equipment failure prediction, achieving 89% precision and 91% recall on the holdout test set
- Created a feature engineering pipeline extracting 200+ time-series features from sensor data, improving model performance by 18%
- Implemented real-time alerting system triggering maintenance schedules, preventing equipment downtime and saving the company $2.3M annually

Multi-Modal AI Agent | 2022

- Built autonomous AI agent using LangChain orchestrating interactions between vision, NLP, and reasoning models for complex task automation
- Integrated with enterprise APIs and knowledge bases, enabling agent to complete multi-step workflows autonomously with 92% success rate
- Reduced manual processing workload by 1200 hours annually, generating $500K in productivity improvements

EDUCATION

Master of Science in Data Science | *University of Texas at Austin* | *Graduation: May 2020*

Relevant Coursework: Advanced Machine Learning, Deep Learning Architectures, Natural Language Processing, Statistical Methods, Cloud Computing for Data Science

CERTIFICATIONS & CREDENTIALS

- AWS Certified Machine Learning – Speciality | *AWS*
- TensorFlow Developer Certificate | *Google*
- Professional Data Engineer Certification | *GCP*
- Machine Learning Engineer Nanodegree | *Udacity*

PUBLICATIONS & SPEAKING

Conference Paper: "Optimization Techniques for Real-Time LLM Inference" | *AI Systems Conference 2024*

Open Source Contributor: LangChain framework (100+ GitHub stars on contributed modules), HuggingFace Model Hub

Technical Blog: 15+ articles on LLM fine-tuning, RAG systems, and production AI best practices (500K+ total views)

ADDITIONAL INFORMATION

Languages: English (Native), Spanish (Professional Working Proficiency)

Technical Interests: Autonomous AI Agents, Prompt Engineering at Scale, Responsible AI & Bias Mitigation, Edge AI Deployment, Federated Learning

Awards: Employee of the Year (2023), Innovation Award for RAG System Architecture (2024)